Nonlinear Calculus in Practice: The Matrix of Error and Cross-System Integration

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Abstract

The Matrix-of-Error framework organizes recoverable physical contributions across dynamical systems. Its earlier development supplied explicit integral calculations, numerical recovery of established equations, and a full three-dimensional Navier–Stokes representation. This paper provides the practical explanation between those statements: how to choose a weak measurement, transfer derivatives by integration by parts, assemble signed contribution columns, recover coefficients, and interpret the resulting matrix. A worked Burgers calculation illustrates the shared integration procedure. Its application to three-dimensional Navier–Stokes is then explained through the vorticity features, full block contraction, and exact enstrophy balance. The continuation envelope is then obtained by its interpolation, absorption, and integrating-factor steps. Connections to the binary-probability series and the density/relativity studies are explained through their actual calculations and motivating relationships. The exposition distinguishes changes of representation, physical interpretation, and identification of an underlying mechanism. Riemann-related mathematics, quantum mass-gap questions, and light propagation are presented as potential applications. The contribution is a reproducible methods bridge and a basis for communicating the completed research, not a new numerical experiment or a replacement for the established mathematics it recovers.

1 From a Stated Result to a Usable Calculation

The earlier nonlinear-calculus paper independently recovered established physical relations across an orbital model, Burgers' equation, and two- and three-dimensional fluid systems [4]. It also supplied the full Navier–Stokes Matrix of Error and its continuation interpretation. Those calculations already exist. The purpose here is to make their execution and meaning explicit enough that a reader can reconstruct the process rather than recognize only its outcome.

Nonlinear Calculus is the broader integration perspective: non-equilibrium physical contributions are expected parts of a model, and established nonlinear uses of calculus belong within that perspective. The *Matrix of Error* is the cross-system organization of contributions connecting a nominal description to the fuller relation. It does not mean that physical behavior is erroneous or that previous nonlinear integration did not exist. *Omni-Dimensional Equilibrium* supplies the proposed zero-error reference cases, while the actual meaning of zero depends on which mathematical object is zero.

The distinction matters in teaching. A coefficient can be absent, nonzero columns can cancel, or the complete equation can close while the state continues evolving. A useful explanation must say which of these is being calculated.

The preceding studies are connected by more than presentation. The unified space-time interpretation motivated the treatment of physical corrections; the matrix program tested that treatment against known systems; the continuation note clarified a resulting analytical condition; and the density study made a specified interpretation computable through a stress-to-metric-to-propagation chain [4, 5, 6, 7]. The binary series develops another explicit representation bridge, from fair records to measure and observation convergence [1, 2, 3]. These connections do not require every paper to use the same operation. Their individual mathematical roles are part of the explanation.

The interpretation developed in the relativity study [7] takes existing relativity as its starting point. Its guiding question is *how are things relative?*: how mass, motion, gravity, light propagation, and temporal relationships fit within one interconnected spacetime account. In this usage, *extended* refers to pursuing that interpretation through its explanatory connections and applications, not to proposing a modification of relativity's physical laws. Separating quantities for calculation does not mean treating their physical relationships as unrelated. This primer makes the reasoning behind those connections explicit, shows how it motivated the completed calculations, and identifies the further questions they make possible.

2 The Shared Weak Calculation

2.1 Inputs, boundaries, and outputs

Suppose a scalar component q is represented by

$$\partial_t q = \sum_{j=1}^{m} \xi_j \mathcal{F}_j(q). \tag{1}$$

The supplied terms encode the system's candidate contributions. Different systems have different variables and terms; sameness of the integration process is not sameness of its inputs. Generating candidate libraries independently of physical knowledge was not the objective of the earlier program.

Choose a test function φ on a space-time cell K. For the time derivative,

$$\int_K q_t \varphi = \left[\int_{\Omega_c} q\varphi \, dx \right]_{t_0}^{t_1} - \int_K q\varphi_t. \tag{2}$$

Compact temporal support removes the endpoint term. For a spatial first derivative,

$$\int_{\Omega_c} (\partial_j f)\varphi = \int_{\partial\Omega_c} f\varphi n_j \, dS - \int_{\Omega_c} f\partial_j \varphi.$$

Compact support, or appropriate matching periodic traces, removes the spatial boundary contribution. A second derivative requires the additional endpoint condition visible in

$$\int_{\Omega_c} \varphi \Delta q = \int_{\Omega_c} q\Delta\varphi + \int_{\partial\Omega_c} (\varphi\partial_n q - q\partial_n \varphi) \, dS. \tag{3}$$

Only after these checks may the signed multi-index rule be used: $\int (D^\beta f)\varphi = (-1)^{|\beta|} \int f D^\beta \varphi$.

The saved implementation uses products of compact polynomial bumps $b(s) = (1 - s^2)^6$ on $|s| < 1$, zero outside. Compact support must apply at the cell boundary, not merely somewhere inside the computational domain. Scaled derivatives include h^{-k} for cell half-width h. The grid order is $[x_1, \ldots, x_d, t]$, with time last. The numerical integration routine evaluates $\int f D^\beta \varphi$; the code using that routine applies the integration-by-parts factor $(-1)^{|\beta|}$.

Weak-form identification across ODEs and PDEs is established in WSINDy, and fluid-equation recovery through weak empirical relations is established in SPIDER [8, 9, 10]. The contribution of the present program is the Matrix-of-Error organization of recoverable physical contributions within a shared cross-system integration and checking process, corroborated by independent recovery of established results. This primer explains how to carry out that process; it does not claim to replace the underlying integration identities or the equations recovered.

2.2 Burgers: compute every column

Write $u_t = \alpha u u_x + \nu u_{xx}$, with the reference $\alpha = -1$. Multiplication by φ gives an integrated equation; (2) supplies its target. The nonlinear term must first be put into flux form: $u u_x = \partial_x(u^2/2)$. The complete weak row is

$$\begin{aligned} y &= -\int_K u\varphi_t, \\ \Theta_1 &= -\int_K \tfrac{1}{2} u^2 \varphi_x, \qquad \Theta_2 = \int_K u\varphi_{xx}, \\ y &= \alpha\Theta_1 + \nu\Theta_2. \end{aligned} \tag{4}$$

Two integrations by parts give the positive diffusion feature. The negative sign in Θ_1 and the negative physical coefficient α must not be combined twice or omitted. The sign can be followed without relying on a verbal convention:

$$\begin{aligned} \text{physical term:} \quad & +\alpha u u_x = \alpha\, \partial_x(u^2/2), \\ \text{integrated term:} \quad & \alpha \int_K \partial_x(u^2/2)\varphi = -\alpha \int_K \tfrac{1}{2} u^2 \varphi_x = \alpha\Theta_1. \end{aligned}$$

Here the spatial boundary term vanishes under the stated support conditions. The column already contains the integration-by-parts minus sign; its fitted coefficient is α itself. For the reference $\alpha = -1$, the contribution is therefore $-\Theta_1$.

Repeat the integrated measurement over cells, obtaining $y = \Theta\xi + r$. A least-squares calculation estimates ξ. With extra prescribed terms, sparse selection can compare subsets of candidate equation terms having nonzero coefficients. The original nonlinear-calculus study [4] fits nominal and correction terms together; fitting a nominal term first can obscure correlated contributions.

For a hand-calculation illustration, not archived simulation data, take

$$\Theta = \begin{bmatrix} 2 & 4 \\ -1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} -1.8 \\ 1.1 \end{bmatrix}.$$

Adding the first row equation to twice the second gives $8\nu = 0.4$; then $\nu = 0.05, \alpha = -1$. The retained right-hand contribution columns are $\Theta\,\mathrm{diag}(\xi) = \left[\begin{smallmatrix} -2 & 0.2 \\ 1 & 0.1 \end{smallmatrix}\right]$. Their sums reproduce the target while keeping diffusion visible.

The archived Burgers weak-form calculation (Experiment 5c) supplies a separate numerical benchmark [4]. With Cole–Hopf solution data, 600 test cells of half-widths 0.35 in space and 0.25 in time, and seed 20260904, the same procedure retained the advection and diffusion terms at 0, 1, 5, and 10% observational noise. Noise was independent Gaussian perturbation with standard deviation equal to the stated fraction of the sampled field's standard deviation. At 10% noise, the saved coefficients were approximately $\nu = 0.047$ and $\alpha = -0.991$, compared with 0.05 and -1. The two-term structure therefore survived this noise level, with finite coefficient error. The hand calculation above checks the algebra; the archived benchmark additionally tests the integration, numerical fitting, and noise response.

2.3 The matrix and its three zeros

For the homogeneous convention,

$$Q = [y \mid -\Theta], \quad c = (1, \xi^{\mathsf{T}})^{\mathsf{T}}, \quad M = Q\,\mathrm{diag}(c), \quad r = M\mathbf{1}. \tag{5}$$

Select correction columns $\mathcal{K}$, defining $E = M_{:,\mathcal{K}}$ and aggregate $a = E\mathbf{1}$. Then $E = 0$ means the selected columns vanish; $a = 0$ permits cancellation among them; $r = 0$ means the whole relation closes. These statements are not interchangeable. Nor is this test-cell matrix the same object as the pointwise physical matrix introduced below.

Physical interpretation

The columns preserve contributions that might disappear in a single net discrepancy. In Burgers, steepening and smoothing operate together. The framework's practical value is to retain their separate magnitudes and signs before discussing their sum.

2.4 Why the integration is not the whole numerical problem

Weak integration avoids differentiating noisy data in the displayed features; it does not eliminate noise, collinearity, or quadrature error. A nonlinear product can alter noise statistics. Underresolved test functions can give poor second-derivative integrals. Proportional columns prevent unique coefficient identification even in exact arithmetic.

The following sequence makes those checks operational. It separates errors in constructing the weak measurements from errors in selecting or fitting a model.

1. **Check axes, sampling, and support.** Match the field shape to $[x_1, \ldots, x_d, t]$ and the derivative multi-index to that order. For each axis z_n, compute the spacings $d_n = z_{n+1} - z_n$ and their maximum deviation from the first spacing. The numerical integration routine assumes uniform axes; a nonuniform record must not be passed to it as though it were uniform. Record how many samples lie within each test-function support, especially for second derivatives.

2. **Check the boundary terms.** Use the endpoint and surface terms displayed in (2)–(3). Establish their vanishing from the chosen test functions and boundary conditions, or retain and evaluate them. Record any numerical boundary contribution before dropping it.

3. **Check a reference relation where one is available.** For a known-result benchmark, evaluate

$$r_{\mathrm{ref}} = y - \Theta\xi_{\mathrm{ref}}, \qquad \rho_{\mathrm{ref}} = \frac{\|r_{\mathrm{ref}}\|_2}{\|y\|_2}.$$

   A large residual can reveal sign, indexing, or quadrature problems before a fit obscures their source. The reference coefficients are a separate verification input, not fitted targets or a requirement for applying the method to an unknown system. If $y = 0$, report the absolute residual instead of dividing by zero.

4. **Check column identifiability.** Compute $n_j = \|\Theta_{:,j}\|_2$, recording zero or numerically negligible columns separately. For the remaining columns, form $\widetilde{\Theta}_{:,j} = \Theta_{:,j}/n_j$ and inspect its singular values, numerical rank, condition number $\kappa = \sigma_{\max}/\sigma_{\min}$, and maximum absolute pairwise correlation. Declare the rank tolerance. A rank-deficient matrix has no finite full-column condition number; correlations with zero variance are undefined, not evidence of independence. Pairwise correlations do not replace the rank check.

5. **Fit with the stated selection rule.** Estimate $\widehat{\xi}$, record selected terms and the fitted residual $y - \Theta\widehat{\xi}$, and express coefficients in the original column units. If normalized columns were fitted with coefficients $\widehat{\beta}_j$, convert back using $\widehat{\xi}_j = \widehat{\beta}_j/n_j$. Fix the fitting and selection settings before the comparison.

6. **Evaluate held-out measurements.** On data excluded from fitting, compute

$$\rho_{\mathrm{test}} = \frac{\left\| y_{\mathrm{test}} - \Theta_{\mathrm{test}}\widehat{\xi} \right\|_2}{\|y_{\mathrm{test}}\|_2},$$

   again using an absolute residual when the target is zero. Report coefficient errors when reference values are known. A complete held-out trajectory is a stronger separation from the fitted record than another nearby cell on the same trajectory.

These are calculation checks, not a requirement to invent new physics before an established result can be recovered.

The archived program contains concrete examples of why these checks matter (Table 1). Each diagnosis concerns the stated calculation or data representation, not a rejection of the physical contribution being sought.

Table 1: Recorded numerical difficulties and the resulting calculation checks. These are archived findings, not new experiments [4].

Calculation	Located difficulty	Practical consequence
Burgers, Exp. 5	A preliminary linear fit absorbed part of the correlated nonlinear advection into its coefficients.	Fit the nominal and retained nonlinear contributions jointly; a residual after a restricted fit need not isolate the omitted physics.
Burgers, Exp. 5b	Spectral second differentiation multiplies a mode's error amplitude by k^2, strongly amplifying high-wavenumber noise.	Transfer derivatives onto analytic weights where the weak identity allows it; Exp. 5c then retained the correct structure through 10% noise.
Mercury, Exp. 7	The columns x/r^n became nearly parallel along the near-circular orbit, making noisy coefficient separation unstable.	Inspect normalized column geometry; integration alone does not remove collinearity.
Taylor–Green, Exp. 8	Symmetry produced directional rank duplication and trajectory-specific vorticity/diffusion correlations.	Check independent physical information, not only residual size; Exp. 8b used grouped operators and distinct held-out multimode flows.
3D pilot, Exp. 8b	Rounded output times were nonuniform although the weak quadrature assumed uniform spacing; narrow supports also underresolved derivatives.	Compare the reference weak residual with the instantaneous PDE check, then correct sampling and support rather than the physical equation.

3 Changing the System Without Changing the Integration Rule

For Mercury's effective orbital model, let q_i be a position component relative to the Sun, $v_i = \dot{q}_i$ its velocity component, and $r = (\sum_i q_i^2)^{1/2}$ the distance from the Sun. The exponent n indexes candidate inverse-power terms. A component acceleration expanded in q_i/r^n gives

$$y_i = -\int v_i \varphi_t \, dt, \qquad \Theta_n = \int (q_i/r^n)\varphi \, dt.$$

No noisy acceleration need be obtained by differentiating velocity. The earlier amplified-effect calculation and extrapolation checked an established precession scale; it was not new astronomical observation [4].

For a three-dimensional incompressible fluid, $\nabla \cdot u = 0$ and $\nabla \cdot \omega = 0$, so the vorticity equation can be written

$$\partial_t \omega_i = -\sum_j \partial_j (u_j \omega_i) + \sum_j \partial_j (\omega_j u_i) + \nu \Delta \omega_i. \tag{6}$$

The component features are

$$y_i = -\int_K \omega_i \varphi_t, \qquad A_i = -\sum_j \int_K u_j \omega_i \partial_j \varphi,$$
$$H_i = -\sum_j \int_K \omega_j u_i \partial_j \varphi, \quad D_i = \sum_j \int_K \omega_i \partial_{jj} \varphi. \qquad y_i = -A_i + H_i + \nu D_i. \tag{7}$$

Both flux features have the same weak sign, with opposite physical coefficients. Viscosity belongs either in the feature or in its coefficient, not both.

The archived Experiment 8 also passed an embedded planar control through this three-dimensional formulation. For

$$u = (u_1(x, y, t), u_2(x, y, t), 0), \qquad \omega = (0, 0, \omega_3), \qquad (\omega \cdot \nabla)u = \omega_3 \partial_z u = 0,$$

the stretching contribution vanishes. Although the stretching terms remained in the prescribed library, the saved control returned no stretching contribution, with all three corresponding coefficients displayed as 0.000. This checks that the implementation does not introduce stretching where it is absent. Because those feature columns vanish on this control, it does not independently identify a universal stretching coefficient as zero. The subsequent genuinely three-dimensional records test recovery of the nonzero contribution.

Experiment 8b in the original nonlinear-calculus study [4] grouped directions within each physical operator. The saved clean mean coefficients were approximately $(-0.999984, 0.999999, 0.0200011)$, compared with $(-1, 1, 0.02)$. All twelve clean fits retained the three physical terms without decoys; mean held-out relative error was approximately 4.21×10^{-4}. At 5% noise, the small diffusion term was retained in only four of sixty fits, while advection and stretching were retained in all sixty. These archived results show both recovery and its recorded sensitivity, not a new run performed for this exposition.

4 From Full Blocks to an Integrated Physical Balance

4.1 The complete contraction

Let $\omega = \nabla \times u$, $S = (\nabla u + \nabla u^{\mathsf{T}})/2$, and $D/Dt = \partial_t + u \cdot \nabla$. Since the antisymmetric part of ∇u annihilates its own vorticity,

$$\frac{D\omega}{Dt} = S\omega + \nu\Delta\omega = E_{\mathrm{NS}}\mathbf{1}_2, \qquad E_{\mathrm{NS}} = [S\omega \mid \nu\Delta\omega]. \tag{8}$$

Transport is on the left, not discarded. In a fixed orthonormal frame, first isolate direction 3: $\eta = (\omega_1, \omega_2)^{\mathsf{T}} \in \mathbb{R}^2$ and $\zeta = \omega_3 \in \mathbb{R}$. Then

$$\omega = \begin{bmatrix} \eta \\ \zeta \end{bmatrix}, \quad S = \begin{bmatrix} B & b \\ b^{\mathsf{T}} & s \end{bmatrix}, \quad \nu\Delta\omega = \begin{bmatrix} C_2 \\ C_1 \end{bmatrix},$$

where $B \in \mathbb{R}^{2\times 2}$, $b, C_2 \in \mathbb{R}^2$, and $s, C_1 \in \mathbb{R}$, pointwise in space and time. Explicitly, $B = (S_{ij})_{i,j=1,2}$, $b = (S_{13}, S_{23})^{\mathsf{T}}$, and $s = S_{33}$; the diffusion blocks follow the same ordering. The other two splits isolate directions 1 and 2 in turn, retaining the remaining two directions in their original order. When changing frame, transform all three objects consistently:

$$\omega \mapsto R^{\mathsf{T}}\omega, \qquad S \mapsto R^{\mathsf{T}}SR, \qquad \nu\Delta\omega \mapsto R^{\mathsf{T}}\nu\Delta\omega.$$

Then ordinary multiplication gives

$$\begin{aligned} \omega^{\mathsf{T}}E_{\mathrm{NS}}\mathbf{1}_2 = \eta^{\mathsf{T}}B\eta + 2\zeta b^{\mathsf{T}}\eta + s\zeta^2 \\ + \eta^{\mathsf{T}}C_2 + \zeta C_1. \end{aligned} \tag{9}$$

The blocks are field-valued, with compatible contractions producing scalars. The cross-block and isolated-direction stretching terms are part of the ordinary multiplication, not additional physical terms introduced after the contraction. Equation (9) is the full expression to evaluate.

Experiment 9 checked four flows, four frames, and three splits. Its direct stretching integral was $\mathcal{B}_{\text{direct}}(t) = \int \omega^{\mathsf{T}} S\omega \, dx$; $\mathcal{B}_{\text{block}}$ was the spatial integral of the first three terms on the right of (9). The saved comparison used

$$R_{\text{full}} = \frac{\|\mathcal{B}_{\text{direct}} - \mathcal{B}_{\text{block}}\|_{2,t}}{\max(\|\mathcal{B}_{\text{direct}}\|_{2,t}, 10^{-15})}, \qquad \|f\|_{2,t}^2 = \int_0^T |f(t)|^2 \, dt,$$

with trapezoidal time quadrature over the saved record and the stated numerical denominator floor. Its maximum was approximately 3.04×10^{-16}. Thus the number compares spatially integrated stretching histories; it is not an error bound for the entire simulated trajectory. The experiment also checked the diffusion contraction and the full enstrophy balance. These are fixed-form algebraic and numerical checks, not another coefficient fit or an all-data error theorem [4].

4.2 The second use of integration by parts

For smooth unforced periodic flow, put $Y = \int |\omega|^2$. Dot the vorticity equation with ω and integrate. The time term gives $Y'/2$; transport vanishes because

$$\int \omega \cdot (u \cdot \nabla)\omega = \tfrac{1}{2} \int \nabla \cdot (u|\omega|^2) = 0.$$

Diffusion gives

$$\nu \int \omega \cdot \Delta\omega = -\nu \int |\nabla\omega|^2.$$

Thus

$$\frac{1}{2}Y' = \mathcal{B} - \mathcal{C}, \qquad \mathcal{B} = \int \omega^{\mathsf{T}} S\omega, \quad \mathcal{C} = \nu \|\nabla\omega\|_2^2 \geq 0. \tag{10}$$

The negative dissipation here agrees with the positive weak diffusion feature in (7): this calculation integrates once against the field, whereas feature construction integrates twice against a prescribed test function.

Physical interpretation

Stretching can increase total squared local rotation, while viscosity dissipates it in this balance. Instantaneous growth is not itself a singularity. The numerical examples plotted in the original nonlinear-calculus study [4] include transient growth and a subsequent crossing into decay within the same representation.

5 Calculate and Interpret the Continuation Envelope

For a fixed unit direction χ, define $P_\chi = I - \chi\chi^{\mathsf{T}}$. The established two-component estimate and interpolation give [11, 12]

$$|\mathcal{B}| \leq C_\Omega \|P_\chi\omega\|_2 \|\omega\|_4^2 \leq K_\Omega \|P_\chi\omega\|_2 \|\omega\|_2^{1/2} \|\nabla\omega\|_2^{3/2}.$$

Here K_Ω includes the interpolation constant. To perform the absorption explicitly, set

$$h = \|P_\chi \omega\|_2, \qquad X = \|\nabla\omega\|_2^2, \qquad Y = \|\omega\|_2^2, \qquad \varepsilon = (1-\delta)\nu, \quad 0 < \delta < 1.$$

Then $|\mathcal{B}| \le K_\Omega h Y^{1/4} X^{3/4}$. Young's inequality with conjugate exponents $4/3$ and 4 gives

$$\begin{aligned} K_\Omega h Y^{1/4} X^{3/4} &= \left(\frac{4\varepsilon X}{3}\right)^{3/4} \left[\left(\frac{3}{4\varepsilon}\right)^{3/4} K_\Omega h Y^{1/4}\right] \\ &\le \frac{3}{4}\left(\frac{4\varepsilon X}{3}\right) + \frac{1}{4}\left(\frac{3}{4\varepsilon}\right)^3 K_\Omega^4 h^4 Y \\ &= \varepsilon X + \frac{27 K_\Omega^4}{256\varepsilon^3} h^4 Y. \end{aligned}$$

Since $\varepsilon X = (1-\delta)\mathcal{C}$, this is

$$\mathcal{B} \le (1-\delta)\mathcal{C} + g_\chi Y, \qquad g_\chi = C_{\Omega,\delta}\nu^{-3}\|P_\chi\omega\|_2^4, \tag{11}$$

where one admissible constant is $C_{\Omega,\delta} = 27K_\Omega^4/[256(1-\delta)^3]$; no optimality of this constant is claimed. The fourth power comes from raising the second Young factor to power 4, and ν^{-3} comes from ε^{-3}. They are consequences of absorbing the gradient contribution while retaining $\delta\mathcal{C}$, not fitted choices. This is the $(4,2)$ member of the established Chae–Choe criterion, whose identification was already made in the original nonlinear-calculus study [4, 5].

Substitute (11) into (10) and set $G(t) = \int_0^t g_\chi(s)\,\mathrm{d}s$. Then

$$\frac{d}{dt}(e^{-2G}Y) \le -2\delta e^{-2G}\mathcal{C},$$

and integration yields

$$Y(t) + 2\delta\int_0^t e^{2(G(t)-G(s))}\mathcal{C}(s)\,\mathrm{d}s \le Y(0)e^{2G(t)}. \tag{12}$$

Finite cumulative control through the finite horizon at issue supplies the conditional continuation statement in the stated setting. This does not demand a finite integral over all infinite time. For example, constant g has finite accumulation at every finite time despite an infinite total as time tends to infinity.

The smallest nonnegative coefficient needed on a particular trajectory is

$$g_{\mathrm{req}} = \frac{[\mathcal{B} - (1-\delta)\mathcal{C}]_+}{Y} \quad (Y > 0),$$

with zero assigned at zero vorticity. It is not identical to the analytic upper bound g_χ. Nor does a positive remainder necessarily mean growth: the retained-margin threshold is $1-\delta$, while the growth threshold for $\mathcal{B}/\mathcal{C}$ is 1.

Forcing adds $\nabla \times f$ as a physical column and $\int \omega \cdot (\nabla \times f)$ to the budget. A source estimate and integrating factor give an additive term in the envelope while preserving the stretching coefficient's functional form. The changed flow can nevertheless have a different accumulated value. The continuation comparison [5] relates this envelope to the smoothly forced breakdown family reported in OpenAI's *Finite Time Blowup for Navier–Stokes*. Accepting that theorem, continuation through the family's singular horizon is excluded, so the applicable control integral must diverge. This exposition does not independently audit the OpenAI proof or present it as the same construction method.

6 The Connections Beyond Fluids

6.1 Binary records: preserve a measure, then calculate

For fair independent bits, the code $X = \sum_{k\geq 1} b_k 2^{-k}$ sends a length-n cylinder of probability 2^{-n} to a dyadic interval of length 2^{-n}. Extension from these generating sets gives the measure identity

$$\mathbb{E}[f(X)] = \int_0^1 f(x)\,\mathrm{d}x$$

for integrable f. The simple calculation $\mathbb{E}X = \frac{1}{2}\sum 2^{-k} = 1/2$ matches $\int_0^1 x\,\mathrm{d}x$. Finite cylinder mass is already normalized; frequency convergence is a different asymptotic property [1, 2].

Repeated observations introduce dynamics. With likelihoods L_i, posterior odds obey

$$O_{n+1} = O_n \frac{L_1(Y_{n+1})}{L_0(Y_{n+1})}.$$

Logarithms turn this product into accumulating evidence. Identifiability, not normalization alone, drives endpoint convergence in the classical model; the cited repeated quantum nondemolition (QND) measurement theorem supplies its quantum counterpart [3]. Independently repeat the complete measurement process, resetting each run to the same symmetric preparation. The binary endpoints identified in the limit of those runs have equal probabilities and supply the fair record of the bare binary biconditional [1, 3]. This record is distinct from the dependent probe observations within a single run.

This is not the fluid weak formulation. The connection is concrete: specify the mathematical object, preserve it under a useful representation, and calculate what follows rather than replacing the calculation with an intuitive resemblance.

6.2 Density, geometry, and propagation

The gravitational-density study [6] specifies the effective stress, in $c = 1$ units,

$$T_s^{\mu\nu} = \rho_d v^\mu v^\nu - \rho_v g^{\mu\nu}.$$

For flow volume J, conservation gives

$$\dot{\rho} + (\rho + p)\dot{J}/J = 0.$$

Dust has $\rho_d J =$ constant; separately conserved vacuum has $\rho_v =$ constant. Equal initial densities are therefore not enough to identify their future behavior or physical origin. The specified perturbation calculation carries density through a metric constraint to a light-travel integral [6].

For a static zero-shift metric,

$$ds^2 = -N^2c^2dt^2 + d\ell^2, \qquad d\tau = N dt, \qquad \Delta t = \frac{1}{c}\int_\Gamma \frac{d\ell}{N}.$$

The null condition gives $d\ell/d\tau = c$ locally while retaining path-dependent coordinate timing. Local and accumulated observations are reconciled by an explicit calculation, not by treating them as identical measurements.

In the completed dust-plus-vacuum example [6], a is the background scale factor, $D(a)$ the linear density-growth factor, and Φ the dimensionless gravitational potential perturbation. For the same spatial mode, the density-contrast amplitude grows by a factor of 15.5804 between $a_i = 0.05$ and $a_f = 1$, while the potential amplitude becomes 0.779022 of its initial value. The expansion ratio is $a_f/a_i = 20$; the relation between the other two ratios follows from $\Phi \propto D/a$. Allowing the potential to evolve during the selected light transit, instead of holding it at its emission value, reduces the calculated delay by about 0.775%. These are model consequences, not a measured frequency shift or a determination of the density's intrinsic origin.

6.3 What the interpretation contributes

The relativity study states the unified space-time interpretation that motivated the research program and its associated matrix-of-error framework [7]. As clarified in the introduction, that study presents an interpretation of existing relativity, pursued through its explanatory connections and applications, rather than proposing a new physical law. Its purpose is to make the interconnected relationships motivating these calculations explicit. This reading of relativity treats spacetime as one interconnected physical structure, with temporal relationships belonging to its local state and history; the reader is asked to examine what established relativity already implies for those relationships, rather than to look for a modified law. Within this interpretation, motion and the active physical structure of space give rise to time, rather than objects merely evolving inside an independently adjustable temporal container. This is stronger than saying that coordinates are related. An individual object being stationary in a chosen coordinate system does not imply that the interconnected system is without motion.

The proposed intrinsic spatial density concerns what carries a gravitational contribution, not a claim that density explains every effect. The author's intended causal chain is:

> Mass influences the local state of spacetime → that interaction produces gravitational behavior → it affects light propagation and temporal relationships, potentially also affecting conditions for life.

This account concerns interconnected regions of one universe, not alternative realities. *Space causes time* expresses the causal interpretation presented in the relativity study [7]: temporal relationships arise from the local state and history of the interconnected spacetime system, rather than time being independently adjustable. The preceding density-to-metric-to-delay calculation supplies a concrete example of part of this chain. Possible consequences for life remain a question about local physical conditions and accumulated processes, not a conclusion about where life does or does not exist.

The interpretation's methodological consequence was to retain physical departures from nominal models and calculate them. The completed matrix recoveries support that mathematical approach; the density calculation makes one specified interpretation operational without selecting its origin.

The interpretation therefore opens questions about mechanisms, observables, and missing relations. It does not require each future result to confirm it. Distinguishing motivation, calculable consequence, and origin is how the connection becomes informative rather than merely a statement of affinity.

6.4 Physical Summary: Dimensional Progression

The dimensional progression explains why the physical contributions change while the integration-by-parts procedure remains reusable. Each of the following models describes an evolving system within spacetime: the dependence on time is present throughout, rather than being added only when moving to the next dimensional case.

One dimension: steepening and smoothing. In the Burgers example of Section 2.2, the scalar field $u(x,t)$ is transported at a rate depending on its own value. This nonlinear transport can steepen its profile, while viscosity smooths gradients. Equation (4) retains the two effects as signed columns measured over the same space-time cell. Their simultaneous presence is the reason to calculate both contributions: a net change alone does not show how much each contributed.

Two dimensions: transport and diffusion of vorticity. For an unforced planar incompressible flow $u = (u_1(x,y,t), u_2(x,y,t), 0)$, the vorticity has only the perpendicular component $\omega_3 = \partial_x u_2 - \partial_y u_1$. Its equation is

$$\partial_t \omega_3 + u_1 \partial_x \omega_3 + u_2 \partial_y \omega_3 = \nu(\partial_{xx} + \partial_{yy})\omega_3.$$

Transport redistributes this local rotation, and diffusion smooths its gradients. The planar structure removes vortex stretching; it does not remove the motion or make transport and diffusion vanish. The weak calculation therefore retains those contributions while respecting the absence of stretching. The embedded planar control in Section 3 checks this distinction within the three-component implementation.

Three dimensions: stretching and coupled components. In the three-dimensional fluid example, vorticity is a vector field whose direction can interact with variations of velocity along it. The contribution $(\omega \cdot \nabla)u$ can stretch and reorient vorticity, alongside transport and viscous diffusion. The calculation must therefore retain the component couplings that the planar case did not require. Stretching and diffusion can contribute with opposing signs to the integrated balance; their individual values remain physically informative even when their sum is small or zero. The full contraction in Section 4.1 retains these couplings even when they vanished in the planar control. Cancellation in an integrated total does not imply that the contributing blocks are absent.

Across these cases, the shared method is to express the system's contributions, integrate them against controlled test functions, transfer derivatives with the boundary terms accounted for, and assemble the resulting signed columns. The physical system supplies the terms and their relationships. The Matrix of Error preserves those contributions so that recovery of the full relation also explains the departures from a nominal description [4].

Further dimensional progression

Further dimensional progression would include familiar objects interacting within spacetime in additional ways. One prospective direction is a four-dimensional formulation that includes solid deformation, with *warp* denoting the solid's change of shape. Such a system would require its own specified variables, dependencies, physical interactions, and boundary conditions before its contribution columns could be calculated. Navier–Stokes is the completed three-dimensional application here, not the definition or limit of the broader framework. What carries forward is the Matrix-of-Error organization and its integration procedure, rather than the particular Navier–Stokes

equations. In this interpretation, spacetime is a liquid-like temporal container, with temporal relationships arising from the interconnected physical activity within it.

7 Potential Applications

The completed calculations suggest applications in arithmetic and the relationship between quantum fields and propagation. The proposals below state the motivating hypotheses and the quantities through which they could be investigated. They extend the research questions, not the list of established results.

7.1 Prime structure, energy-like states, and half-convergence

The arithmetic half-frequency calculations motivate a more specific question than whether the number 1/2 appears in two settings: could their convergence reflect a structure that also governs prime-related fluctuations and the zeta function?

The proposed interpretation treats primes as possible successive *energy-floor* reference states. Intermediate states would carry an excess relative to those reference states, and the stepwise convergence would follow transitions toward the least available potential at each step. This is a proposed energy-like description of arithmetic, not an assigned physical energy of integers. Its mathematical content would be an independently specified state space, energy functional, and transition rule that explain the role of primes, rather than labeling primes as minima by construction. The motivating analogy with the fluid calculations is a reference balance and its retained excess contributions; a shared mechanism is a question for investigation.

There are concrete mathematical comparisons. For any sign record,

$$\frac{\#\{k \le N : s_k = 1\}}{N} - \frac{1}{2} = \frac{1}{2N} \sum_{k=1}^{N} s_k.$$

Thus the speed of approach to half-frequency records the cancellation in the partial sum. Riemann's critical line concerns the real parts of zeta zeros, not this frequency itself [13]. A known bridge is quantitative cancellation: for the unfiltered Möbius sum,

$$M(x) = \sum_{n \le x} \mu(n), \qquad \text{RH} \iff M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0 \quad [14].$$

This is stronger than half-frequency alone. It also requires tracking the original integer cutoff when comparing with the nonzero-Möbius record analyzed in the arithmetic half-frequency study [2].

An energy-based arithmetic representation has an established comparison in Bost–Connes: a Hamiltonian with energies $\log n$ has partition function $\zeta(\beta)$ for $\beta > 1$, with prime factorization giving additive contributions $\log p$ [15]. That construction does not identify primes as energy minima or prove the proposed half-convergence mechanism. It supplies a precise reference against which a new energy-floor model and its discrepancy predictions could be evaluated.

7.2 Mass gaps, spatial structure, and inferred distance

The questions in this section continue the interpretation of interconnected spacetime described in Section 6.3, which motivated the research program and its associated matrix-of-error framework. The prospective quantum-field investigation concerns relationships among mass, spatial structure, and physical behavior, while propagation provides a connected question about how those relationships manifest in observations. The preceding calculations supply methods and worked examples for approaching these questions without predetermining their answers.

A further hypothesis is that accumulated propagation effects could make some inferred separations larger than the corresponding geometric separations, including along paths whose individual local distortions are small. This broadens the density investigation beyond assigning an intrinsic weight to space: such a contribution remains a possibility, but it is not imposed as the explanation.

The author's motivating picture is one ongoing, potentially infinite universe with recurring events and interconnected local spacetime conditions. Within that picture, redshift and the darkness of the sky are questions to investigate through emission and propagation: what reaches an observer, what changes along the path, and why? The investigation would compare accounts without presupposing either cosmic expansion or its absence. An explanation of sky brightness would need to account for the accumulated radiation, not only the appearance of individual sources.

The research question is whether a specified account of spatial structure and propagation could explain part of the observational discrepancies associated with dark matter or dark energy. These involve different evidence, so a distance effect alone would not settle both. The comparison would identify the distance being inferred—for example, luminosity distance, angular-diameter distance, or path travel time—and calculate how the proposed mechanism changes the associated observation. The sign of that change must come from the calculation; a delay or deflection does not by itself show that an object is closer.

The Yang–Mills mass-gap question concerns a positive gap above the vacuum in the energy spectrum of a specified quantum field theory. One established consequence is decay of connected spatial correlations, providing a genuine link between spectral information and spatial behavior [16]. This suggests investigating a chain from quantum-field correlations to an effective gravitational or electromagnetic response, and then to distance-sensitive observables. The gap alone supplies neither that coupling nor a modified light law.

To avoid circular inference, a propagation model should not first assign distances from the effect under examination and then count its agreement with those same distances as independent support. Independently calibrated observations, shared nuisance parameters, and predictions across timing, frequency, image distortion, and gravitational dynamics would distinguish a physical explanation from a freely adjusted distance assignment.

7.3 Light as information about interactions and its path

Radiation can carry information about an emitting interaction while its subsequent path affects what is measured. Gravitational travel-time delay already supplies an established example of the second part [17]. The proposed extension asks whether additional, specified spatial structure leaves a calculable signature in propagation while remaining consistent with local light measurements. A

fluid-like description of spacetime is a candidate effective picture for this investigation, not yet a specified material with a known constitutive law.

The source-to-observer account has four parts: emission; relative motion, including source rotation; propagation through interconnected spacetime regions; and observation. It asks how source behavior and the journey together affect the received signal. A rotation or path effect must yield its own predicted sign and magnitude; it is not assumed to produce a net redshift in every case.

The two outgoing photons in the Higgs diphoton decay channel illustrate the emission side: they carry information used to reconstruct the event [18]. The broader point is that radiation can report physical events while its journey also contributes to the observation.

The proposed connection to mass-gap research is therefore an explicit question: can a quantum-field model determine spatial correlations and a coupling that together predict a photon-propagation effect? Emission, transport, and detection must each be represented so that a source effect is not mistaken for a path effect. Timing, redshift, deflection, and attenuation provide distinct observables rather than interchangeable descriptions of distortion.

These applications preserve the practical aim of the preceding sections: make the relevant contributions calculable, compare their consequences, and use both agreement and disagreement to sharpen the explanation. Each computational study would have its own stated data, comparison, and resource bounds; no new experiment is reported here.

8 Conclusion

The practical content of the Matrix-of-Error framework can be taught through a continuous calculation: choose physical variables and terms, integrate against controlled weights, preserve signs and boundaries, assemble and check the contribution matrix, then recover the physical interpretation from its complete contraction. In three-dimensional Navier–Stokes, the same explanation reaches the exact enstrophy identity and the explicit conditional envelope. The shared calculation and its known-result checks are already completed; the present contribution is to expose their working in a reusable form.

The binary and relativistic studies connect through their own exact representation and propagation calculations, and through the questions that motivated the program. Their relationship need not be reduced to one formula or promoted into an established common microscopic mechanism. Explained at the level of operations, assumptions, and physical meaning, the connections provide both a coherent presentation of the completed work and a concrete starting point for potential applications.

Preparation

The author's longstanding questions about the mass discrepancy associated with dark matter led to investigating relativity through the interpretation of interconnected regions of spacetime described in Section 6.3. This perspective motivated the research programme, including the Matrix-of-Error calculations and the parallel investigations of infinite behavior in quantum states. The Yang–Mills

mass gap entered later as a related but distinct quantum-field question. Propagation connects these stages by asking how spacetime relationships manifest in what is observed.

The author originated the research perspective, directed the prior program and its interpretation, and requested this methods bridge. This exposition was assembled with Codex assistance from the completed papers, saved calculations, and the author's subsequent explanations. The numerical results quoted here are archived results; the small arithmetic illustrations are teaching calculations. No research experiment was rerun and no earlier manuscript was revised.

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